MSG No. 117.305 $P_C\bar{4}b2$ [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'm'2`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4m'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,5,6,11,12,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4,7,8,9,10,13,14]`

Table 3: Wyckoff site: `2c`, site symmetry: `-4'm'2`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[5,6,7,8,9,10,11,12]`

Table 4: Wyckoff site: `2d`, site symmetry: `-4m'2'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,5,6,11,12,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,4,7,8,9,10,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$	`[1,3,10,12]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$	`[2,4,9,11]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ 0\right]$	`[5,8,14,15]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$	`[6,7,13,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[1,3,10,12]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[2,4,9,11]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[5,8,14,15]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[6,7,13,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `2m'm'`.

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,15,16]`
2	$\left[0,\ \frac{1}{2},\ -z\right]$	`[3,4,13,14]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{2},\ 0,\ z\right]$	`[7,8,9,10]`

Table 8: Wyckoff site: `4h`, site symmetry: `2m'm'`.

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[3,4,13,14]`
3	$[0,\ 0,\ -z]$	`[5,6,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[7,8,9,10]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,3]`
2	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[2,4]`
3	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[5,8]`
4	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[6,7]`
5	$\left[x+\frac{1}{2},\ x,\ 0\right]$	`[9,11]`
6	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	`[10,12]`
7	$\left[x,\ \frac{1}{2}-x,\ 0\right]$	`[13,16]`
8	$\left[-x,\ x+\frac{1}{2},\ 0\right]$	`[14,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	`[1,11]`
2	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[2,12]`
3	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[3,9]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[4,10]`
5	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[5,16]`
6	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[6,15]`
7	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,14]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[8,13]`

Table 11: Wyckoff site: `8k`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[1,3]`
2	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[2,4]`

continued ...

Table 11

No.	position	mapping
3	$\left[x+\frac{1}{2},\,-x,\,\frac{1}{2}\right]$	[5,8]
4	$\left[\frac{1}{2}-x,\,x,\,\frac{1}{2}\right]$	[6,7]
5	$\left[x+\frac{1}{2},\,x,\,\frac{1}{2}\right]$	[9,11]
6	$\left[\frac{1}{2}-x,\,-x,\,\frac{1}{2}\right]$	[10,12]
7	$\left[x,\,\frac{1}{2}-x,\,\frac{1}{2}\right]$	[13,16]
8	$\left[-x,\,x+\frac{1}{2},\,\frac{1}{2}\right]$	[14,15]

Table 12: Wyckoff site: 8l, site symmetry: ..2'

No.	position	mapping
1	$[x,\,x,\,0]$	[1,11]
2	$[-x,\,-x,\,0]$	[2,12]
3	$\left[x+\frac{1}{2},\,x+\frac{1}{2},\,0\right]$	[3,9]
4	$\left[\frac{1}{2}-x,\,\frac{1}{2}-x,\,0\right]$	[4,10]
5	$[x,\,-x,\,0]$	[5,16]
6	$[-x,\,x,\,0]$	[6,15]
7	$\left[\frac{1}{2}-x,\,x+\frac{1}{2},\,0\right]$	[7,14]
8	$\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,0\right]$	[8,13]

Table 13: Wyckoff site: 8m, site symmetry: 2'..

No.	position	mapping
1	$\left[\frac{3}{4},\,\frac{3}{4},\,z\right]$	[1,10]
2	$\left[\frac{1}{4},\,\frac{1}{4},\,z\right]$	[2,9]
3	$\left[\frac{1}{4},\,\frac{1}{4},\,-z\right]$	[3,12]
4	$\left[\frac{3}{4},\,\frac{3}{4},\,-z\right]$	[4,11]
5	$\left[\frac{3}{4},\,\frac{1}{4},\,-z\right]$	[5,14]
6	$\left[\frac{1}{4},\,\frac{3}{4},\,-z\right]$	[6,13]
7	$\left[\frac{3}{4},\,\frac{1}{4},\,z\right]$	[7,16]
8	$\left[\frac{1}{4},\,\frac{3}{4},\,z\right]$	[8,15]

Table 14: Wyckoff site: 8n, site symmetry: .m'.

No.	position	mapping
1	$[0,\,y,\,z]$	[1,15]
2	$[0,\,-y,\,z]$	[2,16]
3	$\left[y+\frac{1}{2},\,\frac{1}{2},\,-z\right]$	[3,13]
4	$\left[\frac{1}{2}-y,\,\frac{1}{2},\,-z\right]$	[4,14]
5	$[y,\,0,\,-z]$	[5,11]
6	$[-y,\,0,\,-z]$	[6,12]
7	$\left[\frac{1}{2},\,y+\frac{1}{2},\,z\right]$	[7,9]
8	$\left[\frac{1}{2},\,\frac{1}{2}-y,\,z\right]$	[8,10]

Table 15: Wyckoff site: `16o`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-x,\, -y,\, z]$	`[2]`
3	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	`[3]`
4	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	`[4]`
5	$[y,\, -x,\, -z]$	`[5]`
6	$[-y,\, x,\, -z]$	`[6]`
7	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	`[7]`
8	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	`[9]`
10	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	`[10]`
11	$[y,\, x,\, -z]$	`[11]`
12	$[-y,\, -x,\, -z]$	`[12]`
13	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, -z\right]$	`[13]`
14	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, -z\right]$	`[14]`
15	$[-x,\, y,\, z]$	`[15]`
16	$[x,\, -y,\, z]$	`[16]`